Behavioral Responses of an Infant Gorilla to Maternal Separation in a Captive Social Group of Lowland Gorillas (*Gorilla gorilla gorilla*)

MASAYUKI NAKAMICHI
Osaka University
APRIL SILLDORFF and PEGGY SEXTON
Zoological Society of San Diego

ABSTRACT. The present report describes behavioral reactions of a 33-month-old female infant gorilla to maternal separation in a captive social group whose composition is similar to that of wild groups. The infant gorilla responded differently to two types of maternal separation. When she remained alone for a few hours, she cried repeatedly and moved restlessly, indicating protest. Upon reunion and also for a few days thereafter, an intensification of the mother-infant relationship occurred. On the other hand, when she was separated from her mother but remained with the other group members for nine days, she did not display any behaviors that could be characterized as protest, but rather showed clear behavioral depression. During the 9-day separation period, she hardly played with other group members but she developed affiliative relationships with the leading male. After the reunion from the 9-day separation, she spent as much time near her mother as before separation and played with other infants at a level similar to that before separation, while she sometimes displayed behavioral patterns demanding the maintenance of physical contact with her mother. The present findings are compared with other reports in great apes and human children.

Key Words: Western lowland gorilla; Maternal separation; Infant response; Silverback male-infant interaction; Social support.

INTRODUCTION

This report focuses on behavioral responses of an infant female gorilla (*Gorilla gorilla gorilla*) to two types of maternal separation in a captive social group whose composition is similar to that of wild groups. The responses of human children to maternal separation have been reported in detail (BOWLBY, 1973). A protest phase, characterized by crying for the mother and agitation, occurs immediately upon separation from the mother. When separation continues, a despair phase develops, characterized by reduced activity and social withdrawal. In this phase, the infant can establish new social relationships with another person such as a nurse in a nursery. Upon reunion, a detachment phase occurs in which the infant shows rejection and hostility toward the mother. In addition to these three distinct phases, the infant also shows ambivalence following detachment: on the one hand, the infant demands the mother's presence or cries if she leaves, while on the other hand, the infant shows rejecting or hostile behavior toward the mother (BOWLBY, 1973).

Considerable research in macaque monkeys has revealed a biphasic response of infants to maternal separation: agitated movement followed by behavioral depression, which are consistent with the protest and despair phases in human children, respectively (HINDE & SPENCER-BOOTH, 1971; LAUDENSLAGER et al., 1990; BOCCIA et al., 1991). However, monkey infants have not been reported to show a typical detachment response upon reunion with their mothers.

Some experimental studies have reported that, unlike monkey infants, great ape infants show the complete sequence of protest, despair and detachment in response to maternal separation and reunion, as found in human children (CODNER & NADLER, 1984; HOFF et al., 1994). However, these studies were not conducted in age/sex diversified groups whose composition was similar to that of wild groups and which had remained socially stable for years.

The present report is the first to describe behavioral reactions of a female infant gorilla to maternal separation in a captive social group whose composition is similar to that of wild groups: the group includes her mother, a silverback male, blackback males, adult females, and their young offspring. The female infant was observed in two situations of maternal separation: she was separated from her mother and the other group members, and her mother was removed from the group but she remained with the other group members. These two separations were not performed as part of a separation experiment, but rather to provide medical treatment to the infant and her mother, respectively. Therefore, this study was opportunistic and describes the behavioral responses of only one infant gorilla. However, the findings described below are very useful not only for obtaining a better understanding of social relationships in the gorilla but also for successfully managing breeding groups of captive gorillas.

METHODS

Observations were conducted from August to December 1997 in a social group of western lowland gorillas (*Gorilla gorilla gorilla*) at the San Diego Wild Animal Park, California, USA. In September 1997, the study group consisted of 12 gorillas (see Fig. 2 for their names, sexes, and ages). The composition of this group had been stable without births for three years.

The first author (M.N.) collected data on the three female infants chiefly by the focal sampling method (MARTIN & BATESON, 1986) to describe mother-infant relationships in gorillas. Each focal animal session lasted 10 min. He obtained 319 10-min sessions per infant. He observed the group from the public space for a total of 316 hr over 64 days during the 4-month observation period. Descriptive notes were also recorded by A.S. and P.S. while the gorillas were not only in the outdoor enclosure but also in indoor rooms.

Several behavioral categories were noted by the one-zero sampling method at 2-min intervals while the gorillas were in the outdoor enclosure (approximately 1000m^2) (MARTIN & BATESON, 1986). In the present study, however, only the behavioral categories for one (*Ndjia*) of the three female infants who experienced maternal separation during the observation period were analyzed. When the infant was in physical contact with her mother, this was referred to as contact with mother. Social play included play with others such as wrestling and approach-and-withdrawal play. When the infant interacted with group members other than her mother through behavior such as social play, social grooming, being carried, sitting or lying while maintaining body contact, and walking together keeping a distance of less than 3 m, this was referred to as an affiliative social interaction. Inactivity was recorded when *Ndjia* remained sitting or lying without displaying any scoreable activity throughout the 2-min interval. When she remained sitting, while maintaining passive contact with group members, this behavioral category was not scored.

Ndjia, the female infant gorilla, was born from a multiparous, captive-born mother, *Kamilah*, on December 31, 1994. Her father was *Winston*, the silverback of the group. Since she did not have normal use of the left leg due to a fractured growth plate, she received corrective surgery on the leg at 2 yr and 1 month of age.

On September 23, 1997, six weeks after the observations in the present study started, *Ndjia*, 2 yr and 8 months old, was separated from her mother and the other group members and then anesthetized for a medical examination, which revealed that knee motion was improved. After recovering from the anesthesia, she had to remain alone for approximately two hours in one of the indoor rooms (half-day separation).

On October 19, *Kamilah* was separated from her daughter *Ndjia* and other group members to undergo surgery on her abdomen. Severe adhesive among organs and considerable pus were found in her abdomen. Irrigation was performed, but the etiology of these conditions was unknown. After the surgery, she was housed in an indoor room and she could not see or touch any of the other animals for nine days (9-day separation). During this period, *Ndjia* lived with the other group members.

The 4-month observation period was divided into four phases in accordance with the events of maternal separation: pre-separation (5 weeks, 120 10-min sessions), reunion after the half-day separation (2 weeks, 44 10-min sessions), the 9-day separation (36 10-min sessions), and reunion after the 9-day separation (6 weeks, 119 10-min sessions).

In order to describe proximity-relationships among group members, the location of all 12 gorillas in the group was recorded by scan sampling method approximately once each hour. When two animals were within 5 m of each other, this was referred to as proximity between them. One hundred scans and 119 scans were conducted before and after the 9-day separation, respectively. Overt agonistic interactions among group members such as chase were also recorded ad lib.

RESULTS

Mother-infant Interactions

After the examination of her left leg, *Ndjia* had to stay alone in one of the familiar indoor rooms for approximately two hours. During this period, she moved around and emitted vocalizations repeatedly, indicating protest. As soon as her mother *Kamilah* and the group members returned to the indoor rooms, *Ndjia* approached *Kamilah* and nursed. On the following day (Day 1), *Ndjia* spent 80% of her time in contact with her mother (Fig. 1A). However, from Day 4 onwards, the contact values were almost at the similar level for the pre-separation period.

On the first day of the 9-day separation, *Kamilah* was separated from the other group members including *Ndjia* by keeping her in an indoor room while the other group members moved to the outdoor enclosure. *Ndjia* also moved outdoors together with other group members. She was not recorded to emit vocalizations or to move around restlessly, indicating that she did not fall into the protest phase.

Upon reunion after the 9-day separation, *Ndjia* approached her mother *Kamilah* and climbed on her back. *Ndjia* sometimes nursed from *Kamilah* on that day, but she did not seek the continuous presence of her mother near herself. The differences between the mean contact value for the pre-separation period (18.8%) and that after the reunion from the 9-day separation (29.9%) did not reach, but approached the significant level (*t*-test, $t=1.817$, $df=10$, $p<0.10$, Fig. 1A). In the third week after the 9-day separation, *Kamilah*'s appetite was as much as before her operation, indicating the recovery of her physical condition.

During the period from the third to sixth weeks after the 9-day separation, *Ndjia* displayed some behaviors toward her mother that had never been observed before. These behaviors were

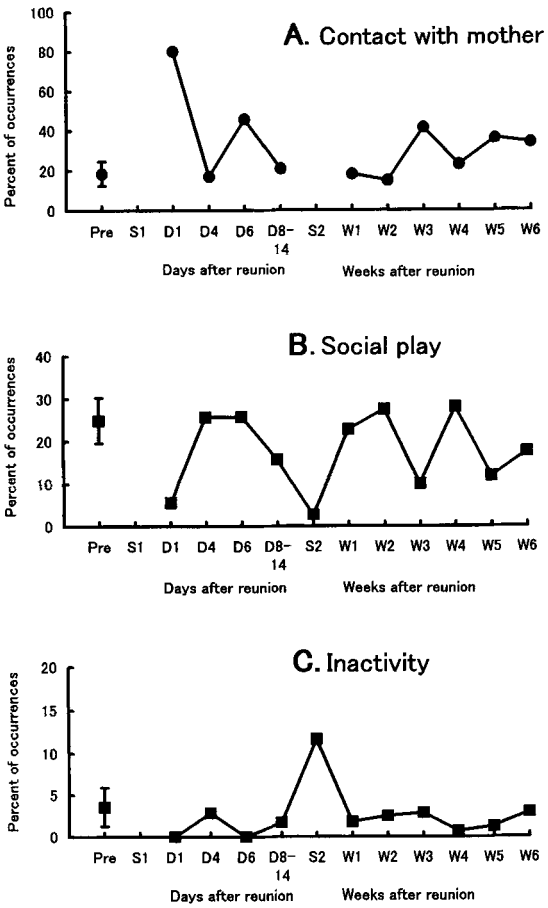


Fig. 1. **A.** Percentage of 2-min intervals during which *Ndjia* was recorded in contact with her mother (*Kamilah*). **B.** Percentage of 2-min intervals during which *Ndjia* engaged in social play. **C.** Percentage of 2-min intervals throughout which *Ndjia* was not recorded to display any scoreable behaviors. Pre: The 6-week pre-separation period; S1: the half-day separation; S2: 9-day separation. Vertical lines at Pre show SD.

each recorded only a few times during the 6-week observation period. When *Kamilah* was sitting on the ground, *Ndjia* leaned on *Kamilah*'s back, putting her hands on *Kamilah*'s shoulder, and then shook *Kamilah*'s back. Directly after this behavior, *Kamilah* started to walk, carrying *Ndjia* on her back. Thus, it seemed that *Ndjia* pressed her mother to walk while carrying her. *Ndjia* was also recorded to try to keep clinging to *Kamilah*'s back when *Kamilah* stopped walking and kept standing on all fours. Thereafter, *Kamilah* shook her back and pushed *Ndjia* with her hand to make *Ndjia* get off. Before the 9-day maternal separation, she always got off her mother's back as soon as her mother stopped walking. *Ndjia* also displayed temper-tantrums: when she was in contact with her mother, she suddenly ran a few meters from her mother, emitting vocalizations. *Ndjia* did not stop crying until she regained body contact with her mother, by being approached by her mother.

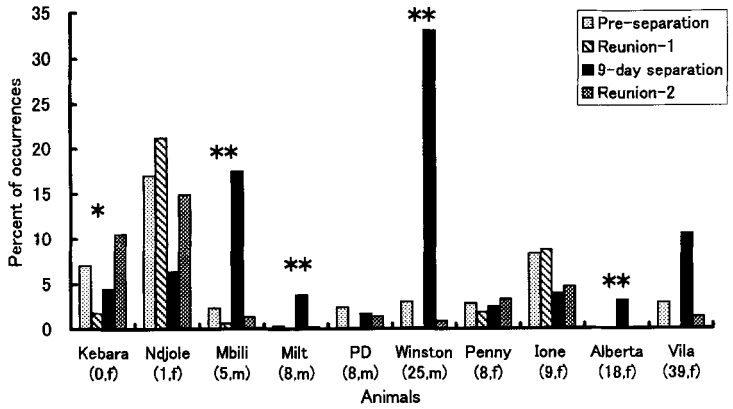


Fig. 2. Percentage of 2-min intervals during which *Ndjia* interacted with group members other than her mother through behaviors such as social play, social grooming, being carried, sitting or lying while maintaining body contact, and walking together keeping a distance of less than 3 m. Figures and f or m in parenthesis show gorillas' ages in year and female or male, respectively. Reunion-1: The 2-week period after the half-day separation; Reunion-2: the 6-week period after the 9-day separation; *Winston*: the leading male and *Ndjia*'s father; PD: *Ndjia*'s older brother. All but the two oldest animals were born in captivity. * $p<0.05$; ** $p<0.01$ for one-way ANOVA. Multiple comparisons by Tukey's method show that for *Mbili, Milt, Winston*, and *Alberta*, the value for the 9-day separation was significantly higher than those for the other three periods, and that for *Kebara* the value for Reunion-1 was significantly lower than that for Reunion-2.

Interactions with Group Members Other than Mother

During the 9-day separation period, *Ndjia* spent only 2.8% of her time in social play (Fig. 1B). She did not respond to play-soliciting behavior such as chest beating in front of her or touching by other infants. After the reunion from the 9-day maternal separation, however, *Ndjia* again started to play with other group members. The mean value of social play for this period (19.7%) was not significantly different from that for the pre-separation period (24.9%) (t-test, $t=1.348$, $df=10$, n.s.).

During the 9-day separation period, the value for *Ndjia*'s inactivity was much higher than that during the pre-separation period, indicating behavioral depression (Fig. 1C).

The frequency of affiliative interactions of *Ndjia* with *Winston*, the silverback male, increased significantly during the 9-day period of maternal separation [ANOVA, $F(3, 16)=18.856$, $p<0.001$, Fig. 2]. She spent 33% of her time in affiliative interactions with *Winston* during the 9-day separation period, while the corresponding values for the other periods were less than 5%. *Ndjia* was often recorded to sit leaning against his back, without displaying any other behaviors. While *Ndjia*'s mother, *Kamilah*, was separated from the group, *Winston* frequently walked around the enclosure, and such frequent walking around was not recorded either before *Kamilah* was separated from the group or after she returned, suggesting that he may have been looking for *Kamilah*. Whenever *Winston* walked around in this manner, *Ndjia* followed him. Since a 5-yr-old male, *Mbili*, also followed him, the value for affiliative interactions with *Mbili* also increased during this period, even though actual interactions accompanied by body contact between them were rarely observed [ANOVA, $F(3, 16)=6.566$, $p<0.001$, Fig. 2].

INTERACTIONS OF *Ndjia*'S MOTHER, *Kamilah*, WITH GROUP MEMBERS

The mean proximity (within 5 m) values of *Kamilah* to *Winston* for 6 weeks before and after the 9-day separation were 43.2% and 35.4%, respectively, and there was no significant difference between them (*t*-test, t=1.076, df=10, n.s.). The corresponding values of *Kamilah* to four adult females were as follows: to *Vila*, 33.5% and 24.3% (t=1.801, n.s.), to *Alberta*, 23.1% and 18.5% (t=0.948, n.s.), to *Ione*, 14.9% and 9.6% (t=1.300, n.s.), and to *Penny*, 10.3% and 16.9% (t=1.553, n.s.). Before and after the 9-day separation period, *Kamilah* was only five and three times, respectively, recorded to have agonistic interactions with other adult group members accompanied with overt aggressive behaviors such as chase. These results indicate that the 9-day separation of *Kamilah* from the group did not produce an over change in her social relationships with group members.

DISCUSSION

CODNER and NADLER (1984) conducted separation experiments on one mother-infant pair of each species of chimpanzee, orangutan, and gorilla, living with only one cage mate. HOFF et al. (1994) also conducted an experiment in which three 27-month-old infant gorillas living with their mothers and a silverback male were separated from their mothers and housed in an unfamiliar cage for 24 weeks. These infants of the two studies showed a protest phase directly after maternal separation. Many researchers have also observed a protest phase for infant macaque monkeys (see Introduction section), like the responses of human infants to separation (FIELD, 1991).

Monkey, gorilla, and human infants in these previous studies had to experience at least one of the following situations: forced separation from the mother by handling, being left alone in unfamiliar physical and/or social environments directly after maternal separation. Like all of these infants, *Ndjia* also protested when she was left alone. On the other hand, *Ndjia* did not show a protest phase during the 9-day separation during which she was allowed to remain in the familiar indoor rooms or outdoor enclosure together with all of the ten remaining familiar group members. The fact that *Ndjia* was not subjected to forced separation by handling or to dramatic changes in the group constellation and physical environments probably related to the absence of a protest phase.

Throughout the 9-day separation period, *Ndjia* hardly played with other group members and spent much more time in inactivity, indicating that she was depressed, like monkey and human infants who are separated from their mothers. During this period, however, she developed affiliative relationships with the silverback male through behaviors such as keeping contact with and following him. In wild mountain gorillas (*Gorilla gorilla beringei*), adult males not only play with juveniles but also hold, carry, and sleep with infants after their mothers have died or transferred to a new troop (FOSSEY, 1983; WATTS & PUSEY, 1993). Therefore, this behavioral tendency of adult male gorillas is exhibited in captivity, and may have helped to reduce *Ndjia*'s psychological stress produced by prolonged maternal separation, although the silverback male did not actually hold or carry *Ndjia*. In other words, how an infant behaves when its mother is removed from the group is greatly dependent on how much social support it can receive. *Ndjia* was able to receive social support from the silverback male. She might have exhibited more depression than actually observed if she could not have received such social support.

Unlike the gorilla infants in the studies cited above and macaque infants (HINDE & SPENCER-

BOOTH, 1971), after the reunion from the 9-day separation, *Ndjia* did not show an increase in stable attachment behavior toward her mother. However, she sometimes displayed some behaviors toward her mother that had never been observed before: shaking her mother's back to press her to walk, remaining clinging to her mother's back while she was standing on all fours, and displaying temper-tantrums. On the other hand, directly after the reunion, the time she spent in social play increased and the time she spent in inactivity decreased. That is, after the reunion, she sometimes demanded particular contact with her mother, while on other occasions, she was able to play with other group members as often as before, without demanding the mother's presence. These findings indicate that, following the reunion, she did not regain the same stable relationship with her mother as before, but her mother's return to the group made it possible for her to regain her active interactions such as social play with other group members. Therefore, *Ndjia*'s psychological state (i.e. being calm or relaxed enough to play with others, but sometimes being upset or demanding particular contact with her mother) seems to be very similar to that of human children who on one hand, demand the mother's presence, while on the other hand, showing rejecting or hostile behaviors toward the mother (BOWLBY, 1973).

The leading male of the present group, *Winston*, walked around very frequently to presumably look for *Ndjia*'s mother while she was absent, but he stopped this behavior pattern as soon as her mother returned. This fact means that the loss of an adult female from a group is likely to influence the behavior of not only her offspring but also that of other group members, particularly the leading male, since the long-term bonds between the leading male and adult females are the social bases of a gorilla group (STEWART & HARCOURT, 1987). Therefore, it is very useful to describe social events which may occur infrequently even in captive groups because analyzing such social events can contribute towards a better understanding of social relationships in gorillas and may be useful for the management of breeding groups of captive gorillas.

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Authors' Names and Addresses: MASAYUKI NAKAMICHI, *Laboratory of Ethological Studies, Faculty of Human Sciences, Osaka University, Suita, Osaka 565-0841, Japan.* e-mail: naka@hus.osaka-u.ac.jp; APRIL SILDORFF and PEGGY SEXTON, *San Diego Wild Animal Park, Zoological Society of San Diego, Escondido, California 92027, U. S. A.*